

The Machine Learning Development Process

Developing a machine learning system involves iterating on both the **model (code)** and the **data**.

- **Model-Centric Approach:** The traditional research approach, where you hold the **data fixed** and iterate on improving the **model/algorithm**.
 - **Data-Centric Approach:** A modern, practical approach where you often hold the **model fixed** and focus on improving the **quality and quantity of the data**. This is often a more efficient way to improve performance.
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Methods for Adding Data

Simply "getting more data" of everything can be slow and expensive. These techniques offer more targeted and efficient ways to improve your dataset.

1. Targeted Data Collection

Instead of gathering all data, use **error analysis** to guide your efforts.

- **Example:** If your spam classifier's biggest problem is "pharma spam," a targeted effort to find and label *only* more pharma spam emails will be far more efficient and effective than just adding more random emails to your dataset.

2. Data Augmentation

This technique creates *new* training examples by applying realistic distortions to your *existing* training examples. The label for the new, distorted example remains the same.

- **Goal:** Tell the algorithm what types of variations are "allowed" without changing the core identity of the data.
- **Key Principle:** The distortions you apply should be **representative** of the types of noise or variations you expect to see in your test set. (e.g., adding random pixel noise to an image is *not* helpful if that's not a realistic scenario).
- **Examples for Images (OCR):**
 - Rotate the image slightly.
 - Enlarge or shrink the image.
 - Change the contrast.
 - Apply random "warping" to the image grid.
- **Examples for Audio (Speech Recognition):**
 - Take the original audio ("What is today's weather?").
 - Add background noise (e.g., a crowd, a car).

- Simulate a bad cell phone connection.

3. Data Synthesis

This is the process of creating *brand new, artificial* data from scratch.

- **Goal:** To generate a massive amount of realistic-looking training data.
 - **Example (Photo OCR - Optical Character Recognition):**
 - **Real Data:** Pictures of text in the real world.
 - **Synthetic Data:** Use your computer to:
 1. Get thousands of different fonts.
 2. Type random characters and words.
 3. Apply random colors, backgrounds, rotations, and distortions.
 - This can generate a nearly infinite supply of training examples that look very similar to real-world text.
 - **Use Cases:** Data synthesis is most common and effective in **computer vision** tasks.
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When you can't easily get or create more data, a technique called **transfer learning** can be a powerful alternative.